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GOING VIRAL: LLM-BASED MODELING OF ECONOMIC NARRATIVES

Abstract. We propose a leakage-safe temporal framework for detecting economic narratives in Russian Telegram news streams. We formulate the task as binary classification on 30K posts (from 123K raw messages) and evaluate it on a Golden Set ($N=500$, $\kappa = 0.794$). Temporal RAG aggregates a 30-day mention history and improves recall ($0.719 \rightarrow 0.912$) and F_2 ($0.697 \rightarrow 0.852$). In contrast, manually engineered diffusion statistics and graph-derived structural features do not yield consistent gains. This suggests narrative formation is semantic and context-driven, and LLMs benefit more from temporal text than aggregated numeric signals. LLM-as-Judge evaluation confirms stable rankings across model families, while human annotations outperform all LLM classifiers.

§1. Introduction

Economic news in social media rarely spreads as isolated updates. Instead, some messages acquire narrative properties: simplified causal explanations, salient consequences, and interpretable economic storylines influencing expectations and behavior.

In Telegram news streams, narratives emerge across channels, reappear in paraphrased forms, and accumulate engagement over time. This makes narrative detection different from topic or sentiment classification: narratives are cross-document, temporally evolving phenomena requiring historical context.

Detecting economic narratives matters because they shape expectations, amplify in digital ecosystems, and may trigger destabilizing cascades.

We model narrative detection as binary classification and propose a leakage-safe pipeline with Temporal RAG and structural signals.

Temporal semantic context proves crucial: injecting a 30-day mention history increases recall from 0.719 to 0.912, improves F_2 from 0.697 to

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0.852, and raises Cohen’s κ from 0.473 to 0.631. In contrast, numeric diffusion features and graph-derived structural signals do not consistently improve performance, suggesting that narrative formation in this domain is primarily semantic and context-driven.

Understanding and tracking economic narratives is important for practical applications such as macroeconomic monitoring, early detection of market sentiment shifts, and policy analysis. Timely identification of emerging narratives allows analysts to capture evolving public interpretations of economic events and assess their potential impact on decision-making and financial markets.

Contributions. Our contributions are as follows:

- We introduce the Economic Telegram News Corpus (30K posts from 123K messages) and a manually annotated Golden Set ($N=500$).
- We formalize economic narrative detection as a temporal classification problem and provide an operational annotation protocol with strong inter-annotator agreement ($\kappa = 0.794$).
- We demonstrate that structured temporal text context significantly improves recall-sensitive performance (+26.8% recall; +22.2% F_2), while aggregated numeric and graph-based structural features do not yield consistent gains.
- We design a leakage-safe Temporal RAG pipeline integrating hybrid retrieval (FAISS + BM25) and temporal ranking for cross-document evidence aggregation.
- We analyze narrative diffusion, showing higher virality and a strong association with negative framing.

§2. Related Work

Shiller formalized narratives as contagious stories shaping economic behavior [29]; later work frames them as interpretative structures assigning causality and responsibility.

Recent studies operationalize economic narratives in large text corpora and link them to financial indicators or expectations [15, 28]. Survey-based analyses show that macroeconomic narratives structure household expectations [3], while media-based narrative indicators have been used to model diffusion dynamics [35]. Most approaches rely on static corpora or econometric modeling.

In NLP, narratives are treated as cross-document semantic structures. RELATIO [4] extracts latent relational patterns from large corpora, and

CANarEx [2] identifies micro-narratives aggregated into temporal clusters. Hybrid models combining Dynamic Topic Models and LLMs have been proposed for detecting narrative shifts [19].

Computational framing research further shows that interpretative frames extend beyond topics to include causality and evaluation [23]. However, most approaches focus on semantic or temporal structure without explicitly modeling behavioral dissemination.

Story detection in news streams originates from Topic Detection and Tracking (TDT) [1, 36], while Dynamic Topic Models introduced probabilistic temporal evolution [6]. Recent neural methods integrate semantic embeddings with temporal-aware clustering for real-time event grouping [22, 30].

Large language models have advanced text classification and structured extraction [7, 32] and are increasingly used in narrative mining pipelines [31].

Information spreads online through cascades characterized by structural virality [13]. Psychological mechanisms further explain why certain messages become socially transmissible [26]. Telegram-specific datasets such as TGDataset enable analysis of cross-channel forwarding dynamics [18].

Prior work studies semantic clustering, temporal evolution, or diffusion separately. We integrate all three within a unified streaming framework.

§3. Problem Formulation

3.1. Narrative Definition. Let $D = \{d_i\}_{i=1}^N$ denote a stream of Telegram news messages, where each message d_i is associated with timestamp t_i and engagement metadata e_i (views, reposts, reactions).

We define a narrative as a socially reproducible interpretative frame emerging across multiple messages over time.

An economic narrative is a temporally evolving subset of messages $N_k \subset D$ that forms a consistent interpretative framing around an economic event or process.

A subset N_k is considered a narrative if it exhibits:

- (1) Semantic coherence: shared framing (e.g., causal explanation, impact evaluation, responsibility attribution).
- (2) Temporal persistence: reproduction across a non-trivial time interval.
- (3) Social propagation: observable dissemination signals in engagement metadata e_i .

In our operational setting, a narrative is defined as a set of at least N semantically related messages occurring within a time interval of length T , where $N \geq 2$ and T is empirically set to 30 days. This definition ensures that narratives capture persistent and reproducible patterns rather than isolated mentions.

A single message does not constitute a narrative; narratives require persistence and reproducible framing. An event report states what happened, whereas a narrative explains why it matters for economic actors.

3.2. Task Definition. While narratives exist at the level of message subsets N_k , detection is operationalized as a message-level classification task.

Given an incoming message $d_i \in D$, the goal is to determine whether the message participates in an emerging economic narrative:

$$f(d_i, C_{t_i}) \rightarrow \{0, 1\},$$

where $C_{t_i} = \{d_j \mid t_j < t_i\}$ denotes temporally preceding contextual evidence.

In practice, the context C_{t_i} is approximated using a temporally constrained retrieval mechanism that aggregates semantically related messages from a fixed historical window preceding t_i .

Context is constructed in a leakage-safe manner using only messages available before time t_i .

§4. Dataset

We construct two datasets: the Economic Telegram News Corpus (30K posts) and a Narratives Golden Set (N=500).

The raw collection contains 123,273 Russian-language posts from seven major Telegram channels collected in 2025. Each record includes text, timestamps, channel IDs, and engagement counters; timestamps are normalized to day level (date_day) for temporal modeling.

The economic corpus (30K+ posts) was constructed by: (1) removing duplicates and near-duplicates, (2) excluding non-news content, and (3) selecting posts assigned to economic topics via an LLM-based classifier (macroeconomics, markets, policy, sanctions/geopolitics, trade, etc.). Topic assignment accuracy on the Golden Set is approximately 90%.

To quantify diffusion dynamics, we compute a composite score `viral_final` over a 3-day window after publication:

$$\begin{aligned} \text{viral_final} = & 0.45 \cdot \text{viral_static} \\ & + 0.20 \cdot \text{viral_dynamic} \\ & + 0.35 \cdot \text{viral_ml}. \end{aligned} \quad (1)$$

The Golden Set (N=500) was sampled to ensure coverage across virality levels and topics. Two annotators independently labeled each post (initial overlap=2), with majority voting after adjudication (up to five annotations). Narrative prevalence is 34.2% (171/500). Inter-annotator agreement for `economic_narrative` is strong (Cohen’s $\kappa = 0.794$).

Narrative posts exhibit significantly higher virality (mean 0.624 vs 0.562, $p < 10^{-8}$; Cohen’s $d \approx 0.57$), yet virality alone is insufficient for detection (ROC-AUC ≈ 0.666). Negative economic framing strongly correlates with narrative presence (54% of negative-effect posts are narratives vs 11% among neutral posts).

Summary statistics are provided in Table 1. Illustrative examples are shown in Appendix A.

Metric	Value
Total messages	500
Narrative prevalence	34.2% (171)
Mean <code>viral_final</code>	0.583
<code>Viral_final</code> (narrative)	0.624
<code>Viral_final</code> (non-narrative)	0.562
ROC-AUC (<code>viral_final</code> $\rightarrow$ narrative)	0.666
Cohen’s κ (<code>economic_narrative</code>)	0.794
Spearman ρ (viral vs strength)	0.256
Spearman ρ (viral vs resonance)	0.210
Narratives among negative-effect posts	54%

Table 1. Summary statistics of the Narratives Golden Set (N=500).

Data Availability. The datasets used in this study are publicly available:

- Economic Telegram News Corpus 2025: <https://huggingface.co/datasets/bruwalkk/economic-telegram-news-corpus-2025>

- Economic Narratives Golden Set: <https://huggingface.co/datasets/bruwalkk/economic-narratives-golden-set>

§5. Models

Our framework consists of three components:

- Temporal RAG retrieval for time-aware evidence aggregation,
- economic narrative classification, and
- LLM-based judging for filtering and consistency analysis.

5.1. Temporal RAG Retrieval.

Hybrid retrieval and leakage safety. Temporal RAG combines dense retrieval (FAISS over multilingual E5 embeddings) and sparse retrieval (BM25). Given a query q and the target publication day `anchor_date`, only documents satisfying `date_day ≤ anchor_date` are eligible, ensuring strict leakage safety.

Temporal RRF fusion. Dense, BM25, and temporal recency signals are fused using a weighted Reciprocal Rank Fusion (RRF):

$$\text{score}_{\text{temporal_rrf}}(d) = \frac{w_{\text{dense}}}{k_{\text{rrf}} + \text{rank}_{\text{dense}}(d)} + \frac{w_{\text{bm25}}}{k_{\text{rrf}} + \text{rank}_{\text{bm25}}(d)} + \frac{w_{\text{time}}}{k_{\text{rrf}} + \text{rank}_{\text{time}}(d)}. \quad (2)$$

where d denotes a document, $\text{rank}_{\text{dense}}(d)$ and $\text{rank}_{\text{bm25}}(d)$ are ranking positions, and k_{rrf} is a smoothing constant.

Temporal rank is defined by document age:

$$\begin{aligned} \text{age_days}(d) &= \text{anchor_date} - \text{date_day}(d), \\ \text{rank}_{\text{time}}(d) &= \text{rank}(\text{age_days}(d) \text{ ascending}). \end{aligned}$$

Thus, newer items receive higher contribution.

Optionally, history is restricted by `max_window_days`. Retrieved context is ordered from newer to older to support timeline reconstruction.

Bounded channel and viral biases. To break ties without overriding relevance, we add two small bounded corrections:

$$\text{score}(d) \leftarrow \text{score}_{\text{temporal_rrf}}(d) + w_{\text{channel}} \cdot \text{channel_w}(d), \quad (3)$$

$$\text{score}(d) \leftarrow \text{score}(d) + w_{\text{viral}} \cdot \left(\text{clip}(\text{viral_final}(d), 0, 1) \right)^\gamma, \quad (4)$$

where $\gamma = 5$ emphasizes strongly viral items while keeping the correction bounded.

Deduplication and clustering. To reduce redundancy, we apply coarse deduplication by normalized text and day, and near-duplicate clustering using cosine similarity ≥ 0.95 and $|\Delta \text{date_day}| \leq 1$. From each cluster, top-scoring representatives are retained.

LLM-based filtering and generation. LLMs are used at three stages:

- **Retrieval judge.** Assigns relevance in $\{0, 1, 2\}$ (keep ≥ 1). Model: Qwen2.5-7B-Instruct.
- **Generator.** Produces a compact source-grounded digest from structured context (typically up to 30 days prior). Model: Qwen2.5-32B-Instruct.
- **Summary judge.** Evaluates digest relevance, grounding, factual consistency, timeline correctness, and recency. Model: Qwen2.5-14B-Instruct.

Here the Retrieval Judge component acts as an LLM-based filtering stage that assigns discrete relevance scores in $\{0, 1, 2\}$, where 0 indicates irrelevance, 1 partial relevance, and 2 high relevance. Only documents with scores ≥ 1 are retained for further processing. This step refines the initial retrieval results by incorporating both semantic similarity and temporal context, effectively reducing noise before passing candidates to subsequent stages.

5.2. Graph-based Structural Features (Ablation). We evaluated a graph-augmented variant where diffusion structure contributes an auxiliary signal after Temporal RRF fusion. A lightweight graph over message similarity and channel connectivity provides normalized features including `msg_sim_degree`, `msg_clustering`, `nbr_hist_viral_share`, `ch_pagerank`, and `ch_out_degree`.

Here, `msg_sim_degree` denotes the number of semantic-neighbor links of a message in the similarity graph, and `msg_clustering` denotes the local clustering coefficient of this message. The feature `nbr_hist_viral_share` captures the historical share of viral messages among neighboring messages. At the channel level, `ch_pagerank` is the PageRank score of the source channel in the channel graph, while `ch_out_degree` is the number of outgoing links from this channel.

Graph score:

$$g(d) = \sum_j \alpha_j \tilde{f}_j(d), \quad g^+(d) = \max(0, g(d)). \quad (5)$$

Bounded gated bonus:

$$\text{score}_{\text{final}}(d) = \text{score}_{\text{temporal_rrf}}(d) + w_{\text{graph}} \cdot \tanh(g^+(d)) \cdot \mathbb{I}_{\text{gate}}(d), \quad (6)$$

where $\mathbb{I}_{\text{gate}}(d)$ restricts the bonus to already strong candidates. Graph augmentation did not improve strict retrieval or classification metrics, and is treated as exploratory rather than core.

5.3. Narrative Classification. Economic narrative detection is formulated as binary classification. We evaluate six instruction-tuned LLMs in zero-shot mode using a unified prompt encoding narrative criteria (broad relevance, causal framing, emotional salience, and potential public impact).

Model selection follows three controlled axes:

- scale within the same architecture,
- cross-architecture comparison at similar parameter size, and
- evaluation of compact models for potential production deployment.
- gpt-oss:20b and gpt-oss:120b [5] are used to quantify the effect of model scale within a single architectural family.
- qwen3-vl:235b-instruct [25] represents a large-scale multimodal model evaluated in text-only mode, serving as a high-capacity upper bound.
- gemma3:27b [14] provides a strong mid-scale baseline optimized for natural language understanding.
- qwen3-vl:8b and WeDLM-8B-Instruct [20] allow comparison between two compact 8B architectures. WeDLM-8B employs diffusion-based parallel decoding under causal attention, initialized from Qwen3-8B, enabling controlled evaluation of architectural differences under fixed parameter count. The WeDLM model was deployed locally, allowing throughput measurement.

5.4. LLM Consistency Analysis. To assess robustness of narrative judgments, we analyze cross-model stability using pairwise win rates and Elo-style ratings [11], measuring agreement with expert annotations.

We intentionally select three large, heterogeneous judge models:

- qwen3.5-397b [25], to test potential intra-family bias toward Qwen-based classifiers;

- deepseek-v3.2 [10], representing sparse-attention long-context architectures;
- minimax-m2.5 [21], optimized for reasoning-intensive evaluation tasks.

Using heterogeneous near-SOTA judges reduces architecture-specific bias and strengthens conclusions regarding model-level agreement.

§6. Evaluation Metrics

We evaluate the system at three levels: retrieval for Temporal RAG, narrative classification, and cross-model consistency. All metrics are in $[0, 1]$ (higher is better).

6.1. Retrieval Metrics. For a query set Q , we report ranking metrics at cut-off k : Precision@ k , Recall@ k , Mean Reciprocal Rank (MRR), and nDCG@ k . MRR is defined as

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q},$$

where rank_q is the position of the first relevant item for query q [33]. nDCG@ k is defined as

$$\text{nDCG@}k = \frac{\text{DCG@}k}{\text{IDCG@}k},$$

and accounts for graded relevance with logarithmic discounting [16]. For generated digests, we additionally use rubric-based LLM scoring for relevance and grounding.

6.2. Narrative Classification Metrics. Narrative detection is a binary task. Let TP, FP, TN, FN denote counts of true/false positives/negatives. We report:

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP + FP}, & \text{Recall} &= \frac{TP}{TP + FN}, \\ F_1 &= 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \end{aligned}$$

We follow the standard definition of F_1 [24]. Since missing emerging narratives is costly, we also report

$$F_2 = 5 \cdot \frac{\text{Precision} \cdot \text{Recall}}{4 \cdot \text{Precision} + \text{Recall}},$$

which weights recall higher than precision. We additionally report ROC-AUC [12] and Cohen’s κ [8].

6.3. Consistency Metrics. To assess robustness across evaluators, we report model-expert agreement (Cohen’s κ), pairwise win rates in head-to-head comparisons, and Elo-style ratings derived from those comparisons [11].

§7. Experiments

We evaluate the proposed framework at three levels:

- zero-shot narrative classification,
- temporal context augmentation via RAG,
- cross-model robustness analysis using LLM judges.

All narrative classification experiments are conducted on the Narratives Golden Set (N=500). Temporal RAG retrieval is evaluated under a separate query-based protocol. Model selection prioritizes F_2 alongside F_1 and Cohen’s κ , reflecting the higher cost of missed narratives (false negatives).

7.1. Zero-Shot Narrative Classification. We evaluate six instruction-tuned LLMs in zero-shot mode using a unified prompt encoding narrative criteria. The prompt remains fixed across all models to ensure comparability. The prompt is designed to reflect the annotation guidelines used for dataset labeling (see Appendix B), ensuring alignment between human annotation criteria and model-based classification.

Compared models: gpt-oss:20b, gpt-oss:120b, qwen3-vl:235b-instruct¹, gemma3:27b, qwen3-vl:8b, WeDLM-8B-Instruct.

This setup allows controlled comparison across model scale within a shared architecture, architectural differences at comparable parameter size, and compact models suitable for deployment.

7.2. Temporal Context Augmentation. To evaluate the contribution of temporal evidence, we augment each target message published at time t_0 with context retrieved from the interval $[t_0 - 30, t_0]$.

The choice of a 30-day temporal window reflects a trade-off between capturing narrative persistence and avoiding excessive topic drift. Shorter intervals may fail to capture slowly evolving narratives, while longer intervals risk merging distinct events. Exploring alternative temporal thresholds is left for future work.

We compare three enrichment strategies:

¹We use vision-language Qwen models for their strong text-processing generalization, without leveraging multimodal inputs.

- `llm_context`: compact plain-text timeline of prior related posts;
- `llm_context_features`: aggregated diffusion and spread statistics only;
- `llm_context_features_full`: timeline combined with spread features.

This ablation isolates whether narrative detection benefits primarily from semantic temporal context, structural diffusion signals, or their combination.

7.3. Temporal RAG Retrieval Evaluation. Retrieval quality is evaluated independently using a query-based protocol, where the task is to retrieve historically relevant posts under strict leakage constraints. We compare BM25, dense retrieval, hybrid search, Temporal RRF fusion, and graph-augmented variants.

7.4. LLM-as-Judge Robustness Analysis. To assess stability of narrative judgments across model families, we implement an LLM-as-Judge evaluation framework. Three large heterogeneous evaluators are used: qwen3.5-397b, deepseek-v3.2, and minimax-m2.5.

Pairwise win rates and Elo-style ratings are computed to quantify relative agreement with expert annotations and cross-model consistency.

§8. Results

8.1. Economic Narrative Classification. Table 2 reports zero-shot classification results on the Golden Set (N=500).

Model	Kappa	F1	Precision	Recall	Accuracy
gpt-oss:20b	0.34	0.51	0.66	0.42	0.73
gpt-oss:120b	0.45	0.66	0.59	0.76	0.74
qwen3-vl:235b-instruct	0.47	0.67	0.62	0.72	0.75
gemma3:27b	0.39	0.65	0.52	0.89	0.68
qwen3-vl:8b	0.36	0.49	0.77	0.36	0.75
WeDLM-8B-Instruct	0.18	0.56	0.42	0.87	0.54

Table 2. Quality metrics for classifying economic narratives

The best overall agreement with expert annotations is achieved by qwen3-vl:235b-instruct ($F1 = 0.67$, $\kappa = 0.47$), closely followed by gpt-oss:120b ($F1 = 0.66$, $\kappa = 0.45$). Within the GPT-OSS family, scaling from 20B to 120B parameters improves $F1$ from 0.51 to 0.66 and κ from 0.34 to 0.45, indicating strong scale dependence.

Mid-scale models exhibit different precision–recall trade-offs: gemma3:27b achieves high recall (0.89) but lower precision (0.52), while compact models show either low recall (qwen3-vl:8b, 0.36) or reduced agreement (WeDLM-8B-Instruct, $\kappa = 0.18$).

8.2. Performance of Local Models. Inference speed for locally deployable 8B models is shown in Table 3.

Model	Average Time (s)	Throughput
qwen3-vl:8b	218.22	0.0046
WeDLM-8B-Instruct	10.39	0.0962

Table 3. Performance of local models

WeDLM-8B-Instruct is over $20\times$ faster (10.39s vs 218.22s), but this gain comes with reduced classification quality (Table 2), illustrating the speed–quality trade-off.

8.3. Temporal Context for Narrative Classification. Table 4 presents the main ablation result.

Variant	F1	F2	Precision	Recall	κ
Text-only (base)	0.667	0.697	0.621	0.719	0.473
+ LLM context (30d)	0.775	0.852	0.674	0.912	0.631

Table 4. Effect of 30-day temporal context (Golden Set, $N=500$).

Adding 30-day temporal context substantially improves recall ($0.719 \rightarrow 0.912$), κ ($0.473 \rightarrow 0.631$), and F_2 ($0.697 \rightarrow 0.852$). The large recall gain is particularly important for monitoring scenarios where missing narratives is costly.

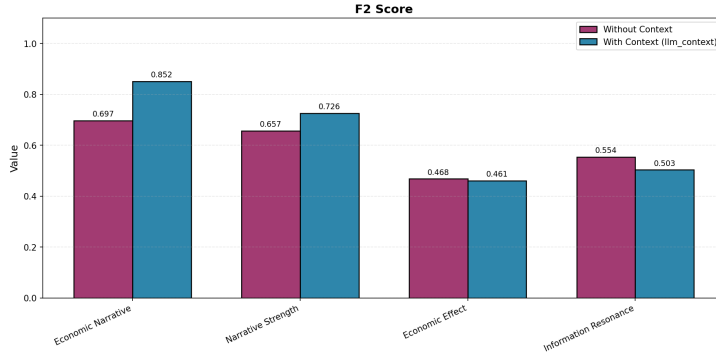


Figure 1. Metric improvements with temporal context augmentation.

Temporal semantic context consistently outperforms purely numeric diffusion features, confirming that structured textual history provides stronger narrative signals.

8.4. Temporal RAG Evaluation. Retrieval quality on 40 queries (top-50) is shown in Table 5.

Method	nDCG@50	MRR@50 (rel=2)	P@50 (rel=2)	P@50 (rel≥1)
BM25	0.739	0.522	0.196	0.385
Hybrid RRF	0.783	0.593	0.268	0.470
Dense (FAISS)	0.763	0.573	0.280	0.484
Temporal RRF (ours)	0.774	0.570	0.266	0.465
Temporal RRF + Graph	0.685	0.412	0.280	0.497

Table 5. Retrieval quality on 40 queries (top-50).

Hybrid retrieval achieves the best ranking quality ($\text{nDCG@50} = 0.783$; $\text{MRR@50} = 0.593$). Temporal RRF maintains competitive relevance while enforcing strict time constraints. Graph augmentation reduces ranking metrics despite slightly higher $\text{P@50 (rel}\geq 1)$, confirming limited benefit of structural signals.

Narrative digest quality (200 queries) is shown in Table 6.

System	Rel.	Ground.	Fact.	Timeline	Recency	Overall
rag	0.857	0.714	0.786	0.714	0.500	0.714
rag_filtered	0.886	0.814	0.857	0.829	0.657	0.809
temporal_filtered	0.900	0.800	0.843	0.843	0.771	0.831

Table 6. Temporal RAG summary quality.

Explicit temporal filtering yields the best overall score (0.831), with strong improvements in recency and timeline correctness.

8.5. LLM-as-a-Judge Consistency. Table 7 reports Elo ratings across three judge models.

Classifier	ELO Range	Win Rate (%)	Matches	Judge Models
gpt-oss:120b	1744–1970	7.6–40	38–201	All 3 judges
gpt-oss:20b	1338–1880	10.4–47	52–235	All 3 judges
qwen3-vl:235b-instruct	1340–1762	1.6–12.1	8–61	All 3 judges
gemma3:27b	1339–1551	1.8–9.3	9–47	All 3 judges
qwen3-vl:8b	1235–1412	0.6–3.4	3–17	All 3 judges
WeDLM-8B-Instruct	1098–1423	0.4–5.2	2–26	All 3 judges

Table 7. Elo ratings for model consistency analysis

Human annotators outperform all LLM classifiers across judges. Among models, gpt-oss:120b shows the strongest judge-based performance. Importantly, no systematic bias toward Qwen-family classifiers is observed from the qwen3.5 judge. Judge variance is moderate: deepseek-v3.2 and qwen3.5-397b show similar patterns, while minimax-m2.5 produces greater differentiation.

8.6. Graph Feature Ablation. Injecting graph-derived structural signals into retrieval ranking or classification prompts does not improve F1, recall, or top- k retrieval metrics. Graph features are therefore treated as exploratory analytical signals rather than core pipeline components.

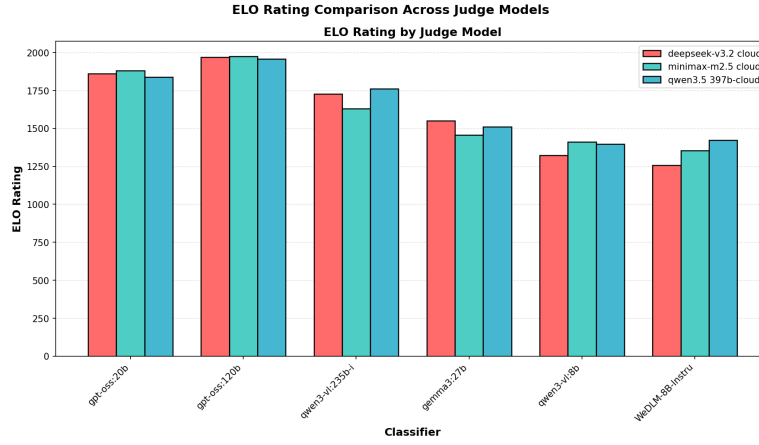


Figure 2. Elo rating comparison across judge models

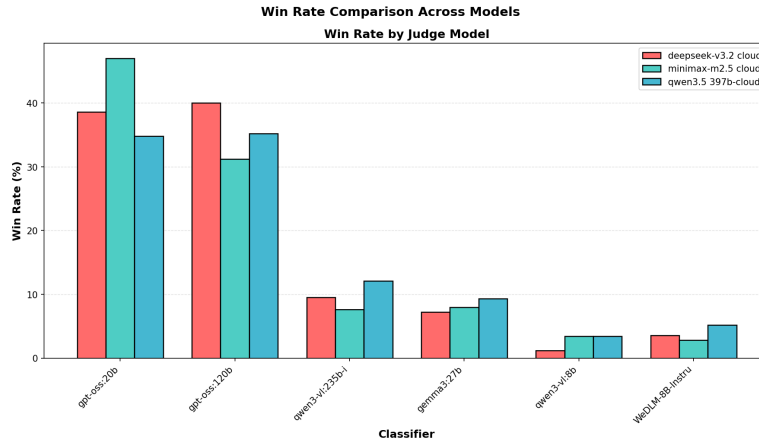


Figure 3. Win rate comparison across models

§9. Discussion and Limitations

Our results show that economic narrative detection is primarily driven by time-aware semantic context rather than manually engineered structural features.

Although diffusion and virality characteristics demonstrate statistically significant correlations with expert narrative labels (Spearman’s ρ up to 0.29; $p < 10^{-8}$), incorporating these numeric features into LLM prompts does not improve classification performance. In contrast, providing structured temporal text context (daily mentions and channel-distributed history) consistently yields higher agreement with manual annotations. Using only aggregated spread statistics performs worse than textual context, and combining both approaches does not produce additional gains.

This suggests an important property of LLM reasoning: models appear to interpret narrative-related propagation patterns more effectively when presented as structured temporal text rather than as abstract numerical summaries. In other words, LLMs demonstrate stronger qualitative temporal reasoning than quantitative aggregation reasoning in this setting.

Several limitations should be acknowledged. First, the dataset covers seven major Russian-language Telegram channels from 2025; generalization to other languages or platforms remains open. Second, the Golden Set contains 500 manually annotated posts; larger labeled datasets would allow more robust validation. Third, classification is evaluated in a zero-shot setting and may be sensitive to prompt formulation and model family. Finally, engagement signals alone are insufficient predictors of narrative presence (ROC-AUC ≈ 0.666), despite their statistical correlation with narrative labels.

Despite these constraints, the proposed framework provides a reproducible and leakage-safe methodology for time-aware economic narrative monitoring.

§10. Conclusion

We formulate economic narrative detection in Telegram news streams as a time-aware classification problem. Unlike topic or sentiment analysis, economic narratives represent socially reproducible interpretative frames that evolve across messages and accumulate diffusion signals.

Empirically, temporal semantic context proves crucial. Injecting a 30-day mention history increases recall from 0.719 to 0.912, improves F_2 from 0.697 to 0.852, and raises Cohen’s κ from 0.473 to 0.631. In contrast, numeric diffusion features and static graph-based signals do not consistently improve performance.

Overall, effective narrative monitoring requires structured temporal semantic modeling rather than purely structural or statistical analysis. Future work may extend this framework toward real-time narrative tracking and integration with macroeconomic forecasting signals.

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Моделирование экономических нарративов с использованием LLM

Аннотация. В работе предлагается временно-ориентированный подход к обнаружению экономических нарративов в новостных потоках Telegram. Задача формализуется как бинарная классификация сообщений с использованием больших языковых моделей. Предлагаемый метод основан на использовании временного контекста (Temporal RAG), агрегирующего историю упоминаний за 30-дневное окно, а также на анализе структурных характеристик распространения информации. Эксперименты на корпусе из 30 тыс. сообщений и вручную размеченном золотом наборе (N=500) показывают, что учет временного текстового контекста существенно повышает полноту и согласованность с экспертными аннотациями, тогда как агрегированные числовые признаки распространения не дают стабильного улучшения качества. Полученные результаты свидетельствуют о том, что формирование экономических нарративов в большей степени определяется семантическим и контекстным содержанием, чем структурными характеристиками диффузии.

Ключевые слова: экономические нарративы, Telegram, большие языковые модели, RAG, временной контекст, анализ текстовых данных

Appendix §A. Illustrative Examples

Message (RU + EN translation)	Label	Rationale
EN: “Mortgage overpayment is enough to buy six one-room apartments in Moscow. With an apartment price of 12 million rubles, the overpayment to the bank amounts to 74 million — six times the original loan. The monthly payment reaches 232,000 rubles.”	Strong narrative	The message contains concrete numerical evidence and frames the issue as a large-scale social burden affecting ordinary people. It establishes a clear causal link between financial policy and everyday economic hardship.
EN: “Rosselkhoz nadzor denied a ban on the import of mandarins from Abkhazia. This information does not correspond to reality. Russia has not introduced any restrictions on imports from Abkhazia,” the agency reported.	Not a narrative	The message is a factual clarification without emotional framing, causal amplification, or broader economic generalization. It does not attempt to construct a socially salient explanatory storyline.

Table 8. Illustrative examples of narrative and non-narrative messages from the Golden Set.

These examples illustrate the operational distinction between fact-reporting and narrative framing.

Appendix §B. Annotation Guidelines (abridged)

Setup. Messages are annotated independently with overlap = 2; annotators are instructed not to inspect each other’s labels. If two initial annotations disagree, we collect up to three additional labels (five total) and aggregate by majority vote.

Labels.

- Economic effect (strong negative / weak negative / neutral / weak positive / strong positive).
- Information resonance (low / medium / high): potential mass-audience salience.

- Topic correctness (agree / partially / disagree).
- Economic narrative (yes/no): whether the post can become a viral explanatory story.
- Narrative strength (low / medium / high): how strongly the post can influence a broad audience.

Narrative checklist. Assess the post from the perspective of an average Russian reader and consider: a clear trigger event, emotional framing (fear/uncertainty/anger/optimism), simplification/generalization (“prices will surge”, “everyone will get poorer”), explicit causal links to economy/markets/everyday life, and references to authorities/experts and broad public discussion.

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